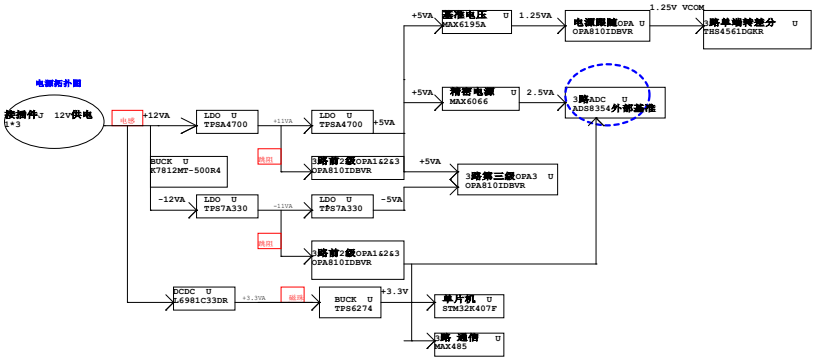
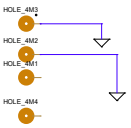
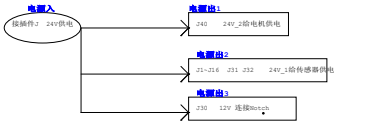
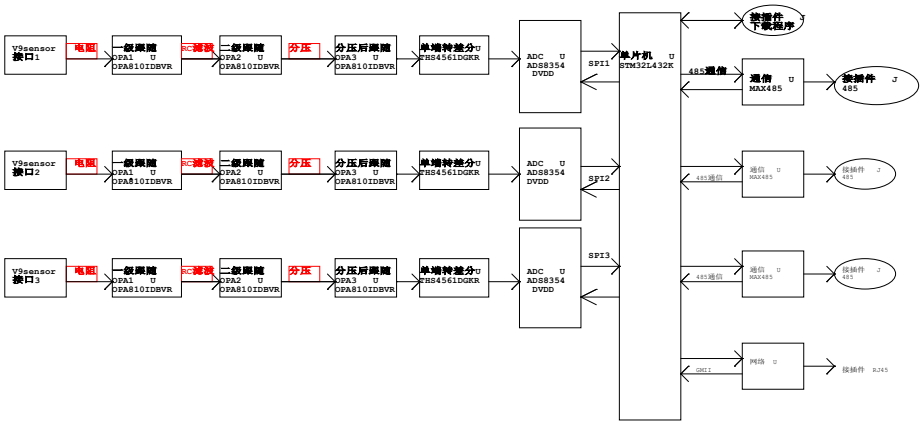




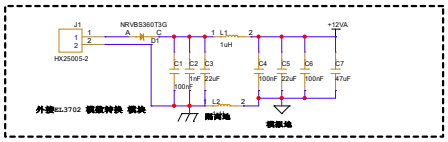


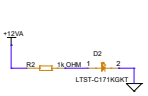
进入板子的24V，在板内分三路 一路24V给电机，一路24V给传感器，一路转成12V给Notch

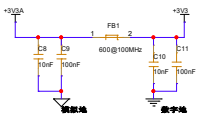
整体方案设计框图



Title	<Title>		
Size C	Document Number <Doc>		Rev <RevCode>
Date: Thursday, December 11, 2025	Sheet 1 of 11		



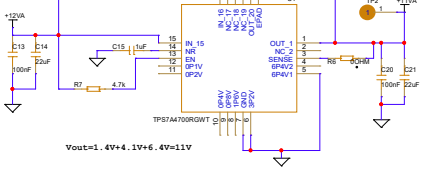




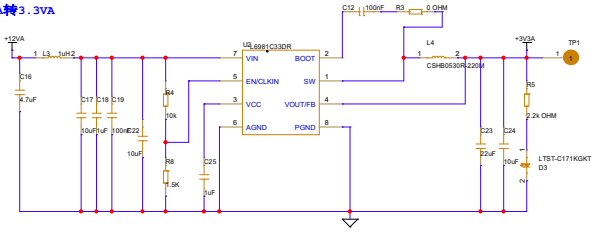


12VA转11VA

IN3~36V OUT1.4~20.5V 1A



12VA转3.3VA



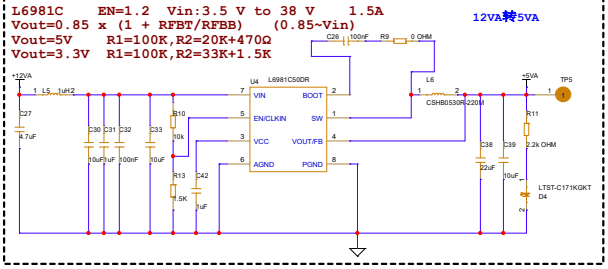
L6981C EN=1.2 Vin:3.5 V to 38 V 1.5A

Vout=0.85 x (1 + RFBT/RFBB) (0.85~Vin)

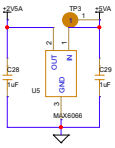
Vout=5V R1=100K,R2=20K+470Ω

Vout=3.3V R1=100K,R2=33K+1.5K

12VA转5VA



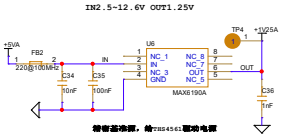
IN2.5~12.6V OUT2.5V



精密基准源 他ADS8354 外部电压参考

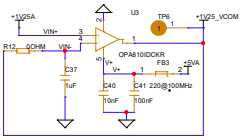
5VA转1.25VA

IN2.5~12.6V OUT1.25V

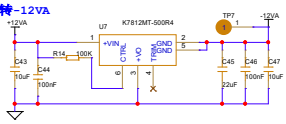


精密基准源，他TPS45611驱动电源

驱动电源跟随：使输出阻抗小，增大驱动能力；保护后端电路；输入阻抗大



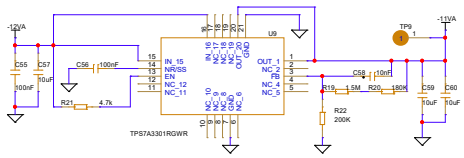
12VA转-12VA



-12VA转-11VA

Vout=(1+R1/R2)Vref=(1+(1.5M+180K)/200K)=11V Vref=1.175V

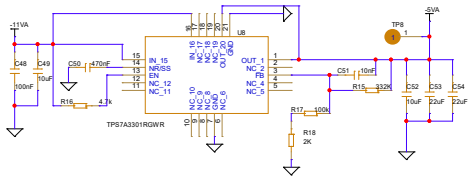
IN-3~-36V OUT-1.18~33V



-11VA转-5VA

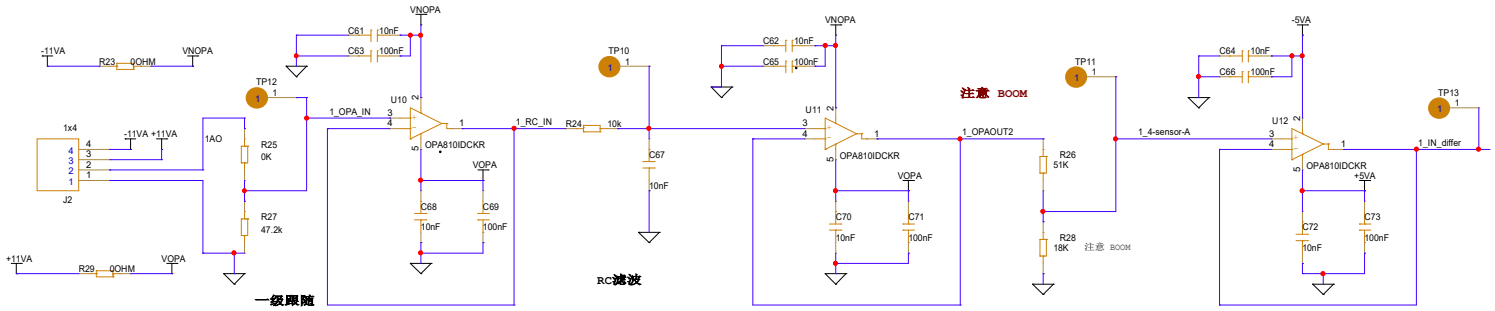
Vout=(1+R1/R2)Vref=(1+(1.5M+180K)/200K)=11V Vref=-1.175V

IN-3~-36V OUT-1.18~33V



Title	<Title>		
Size C	Document Number <Doc>		Rev <RevCode>
Date: Sunday, January 04, 2026	Sheet 2 of 11		

R12 与FB10， C828381选择上件，调节可输入范围



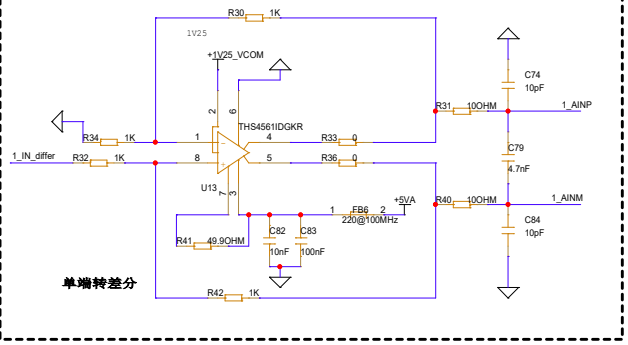
一级跟随

RC滤波

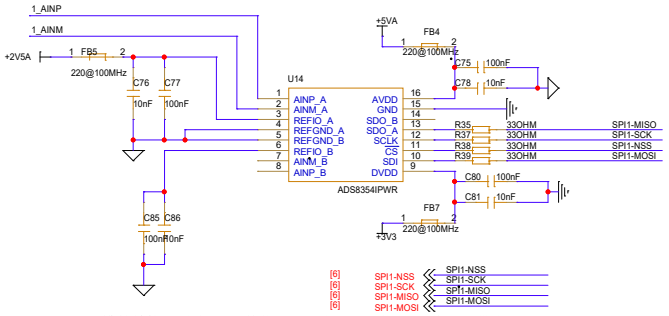
二级跟随

放大后跟随

R25 0 Ω R27不安装

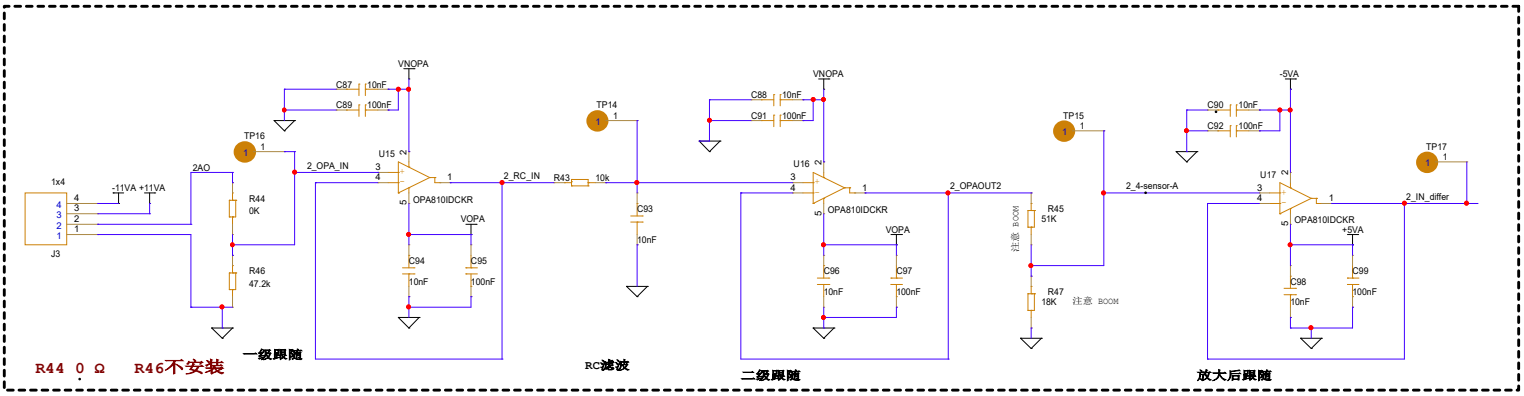


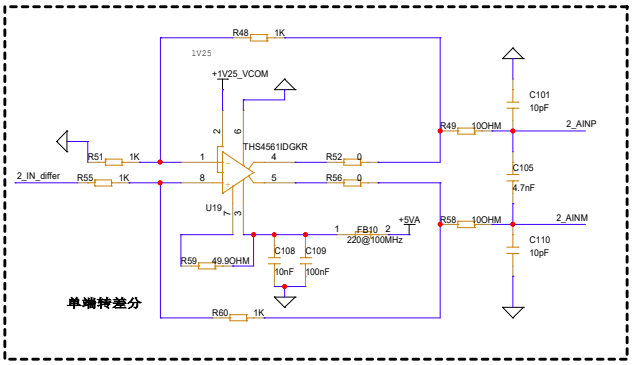
单端转差分

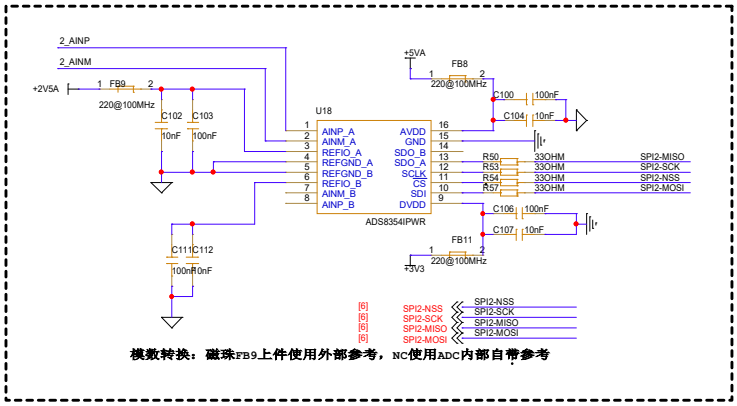


模数转换：磁珠FB7上件使用外部参考，NC使用ADC内部自带参考

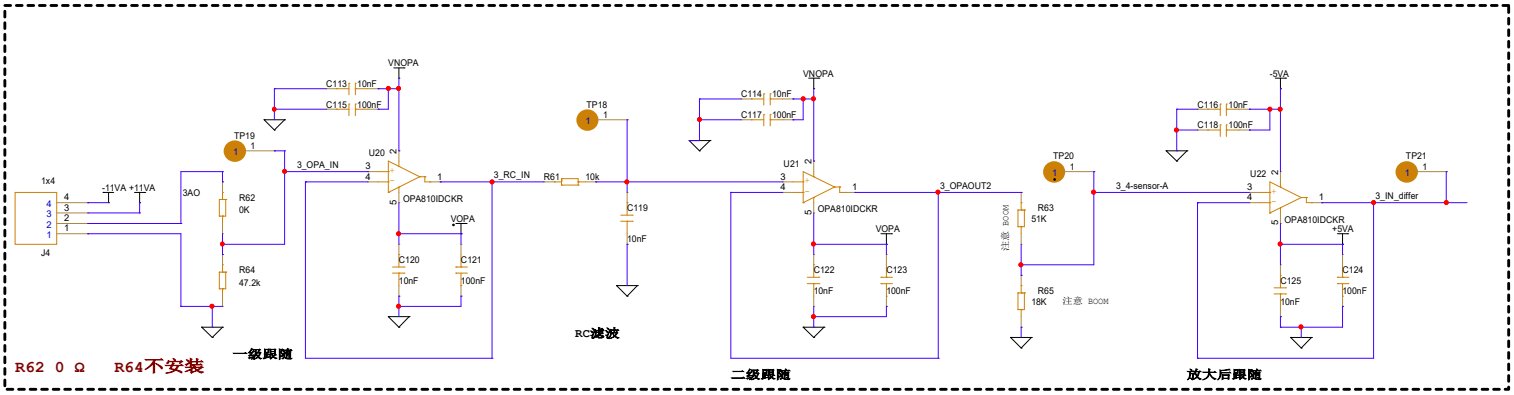
Title	<Title>	
Size B	Document Number <Doc>	Rev <RevCode>
Date: Sunday, January 04, 2026	Sheet 3 of 11	

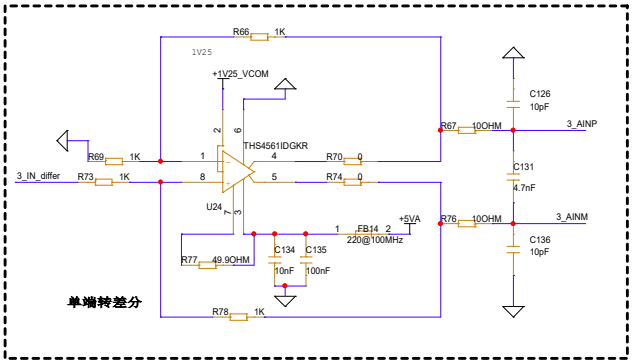


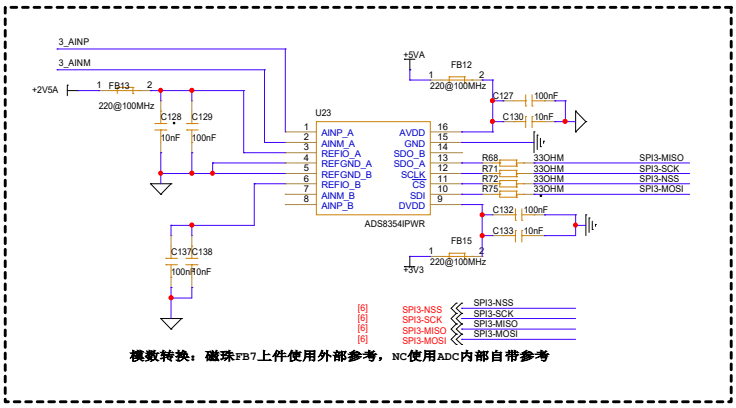




Title	<Title>	
Size B	Document Number <Doc>	Rev <RevCode>
Date: Tuesday, November 18, 2025	Sheet 4 of 11	







Title	<Title>	
Size B	Document Number <Doc>	Rev <RevCode>
Date: Tuesday, November 18, 2025	Sheet 5 of 11	

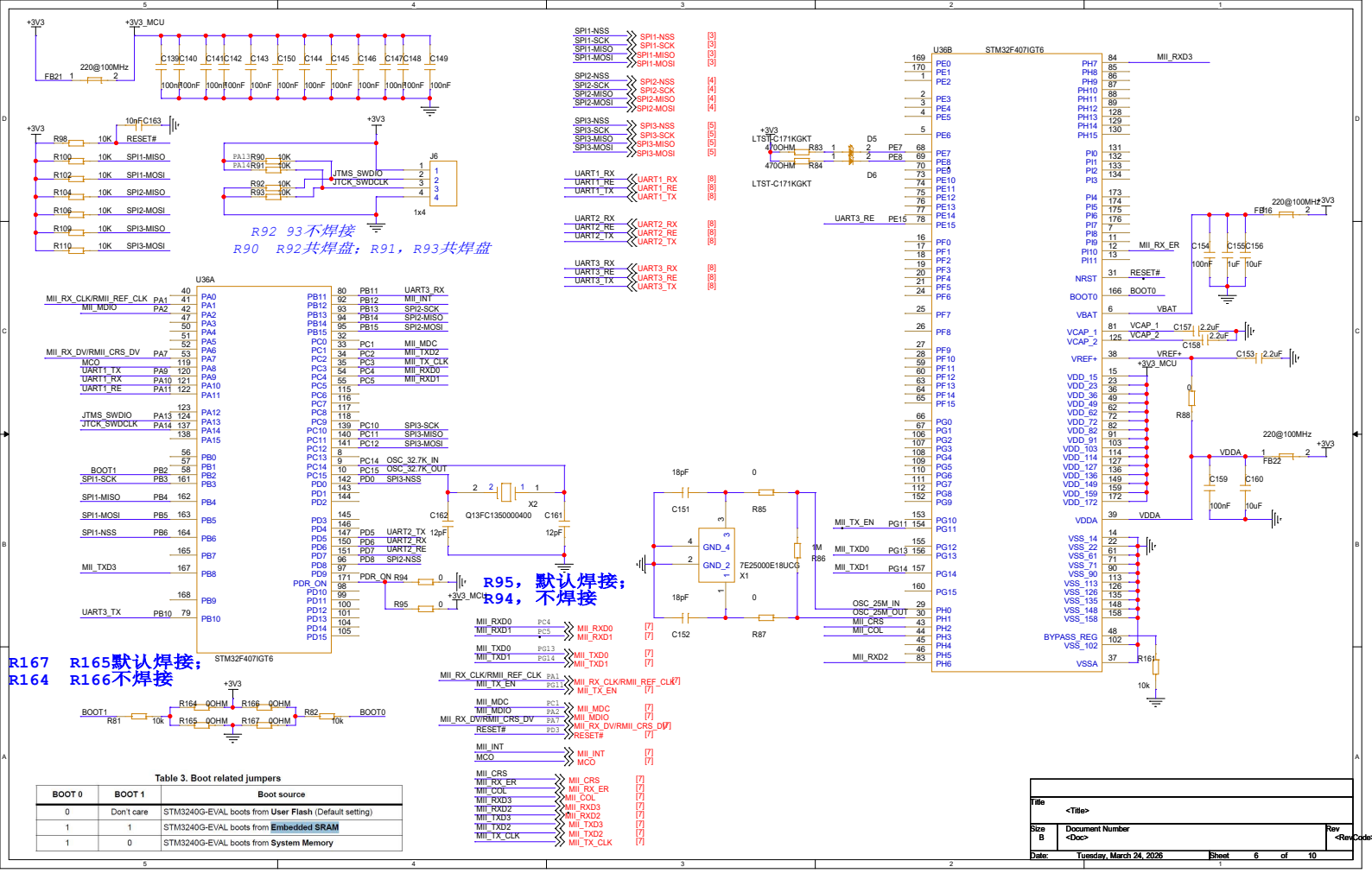
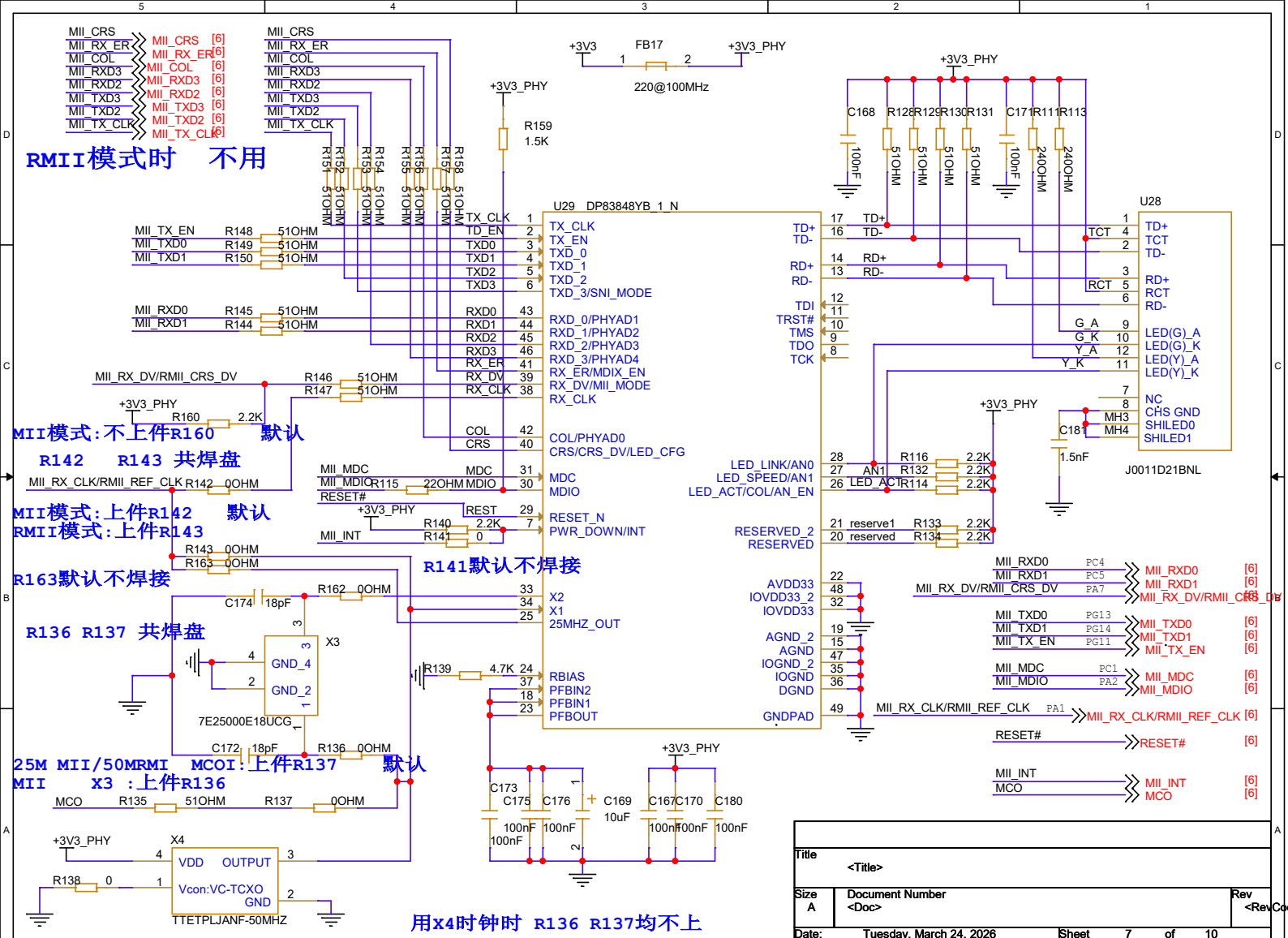
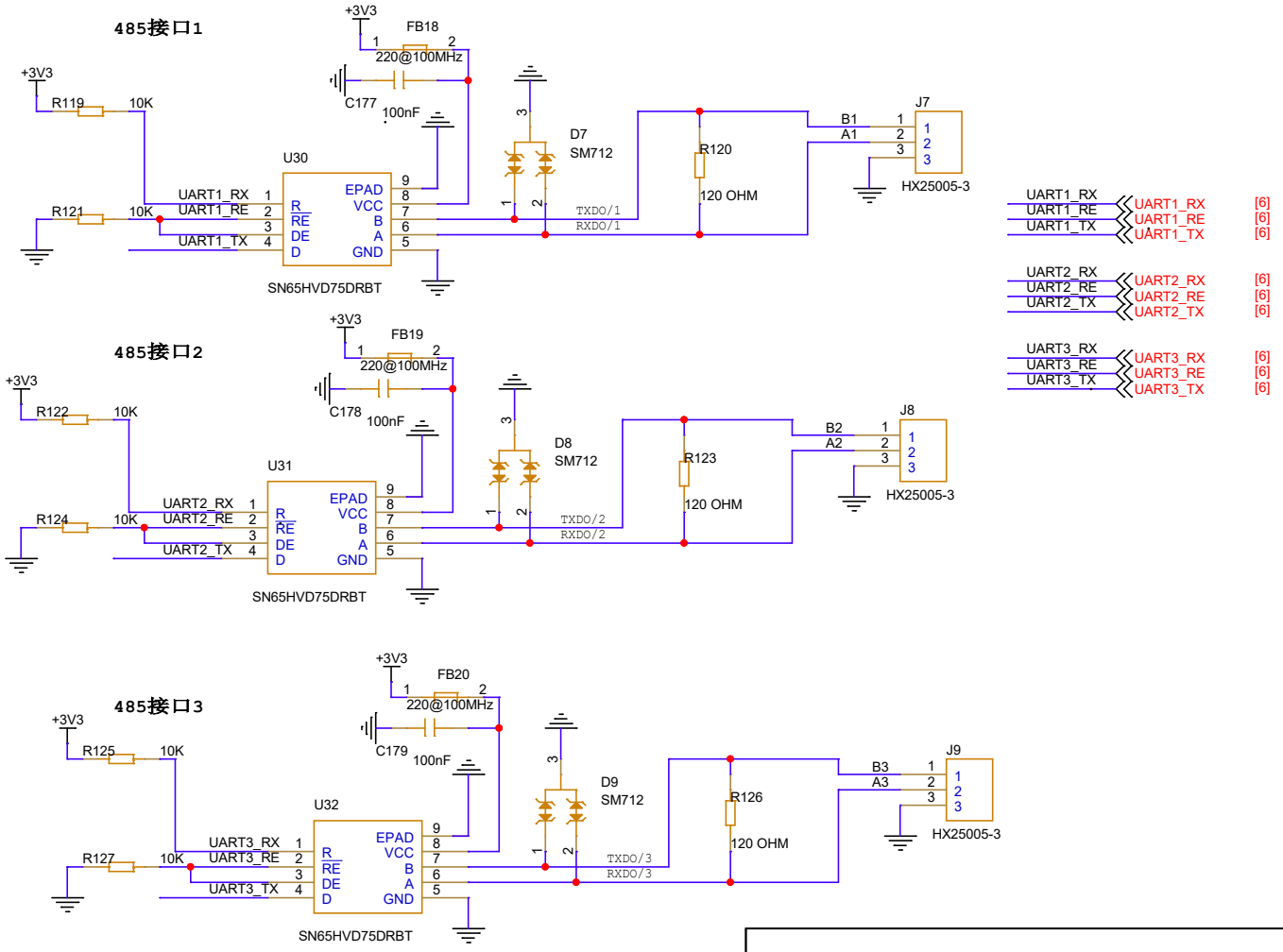


Table 3. Boot related jumpers

BOOT 0	BOOT 1	Boot source
0	Don't care	STM3240G-EVAL boots from **User Flash** (Default setting)
1	1	STM3240G-EVAL boots from **Embedded SRAM**
1	0	STM3240G-EVAL boots from **System Memory**

Title	<Title>	
Size B	Document Number <Doc>	Rev <RevCode>
Date: Tuesday, March 24, 2026	Sheet 6 of 10	


RMII模式时 不用
MII模式:不上件R160 默认
R142 R143 共焊盘
MII模式:上件R142 默认
RMII模式:上件R143
R163默认不焊接
R141默认不焊接
R136 R137 共焊盘
25M MII/50MRMI MCOI:上件R137 默认
MII X3 :上件R136
用X4时钟时 R136 R137均不上
U29 DP83848YB_1_N
U28
J0011D21BNL
FB17
220@100MHz
+3V3_PHY
+3V3
X3
X4
7E25000E18UCG
TTETPLJANF-50MHZ
Title
<Title>
Size A
Document Number
<Doc>
Rev
Date: Tuesday, March 24, 2026
Sheet 7 of 10


485接口1
+3V3
FB18
220@100MHz
C177
100nF
R119
10K
R121
10K
U30
UART1_RX
UART1_RE
UART1_TX
R
RE
DE
D
EPAD
VCC
B
A
GND
SN65HVD75DRBT
D7
SM712
TXDO/1
RXDO/1
R120
120 OHM
B1
A1
J7
HX25005-3
485接口2
FB19
C178
R122
R124
U31
D8
R123
J8
485接口3
FB20
C179
R125
R127
U32
D9
R126
J9
UART1_RX
UART1_RE
UART1_TX
UART2_RX
UART2_RE
UART2_TX
UART3_RX
UART3_RE
UART3_TX
[6]
Title
<Title>
Size
A
Document Number
<Doc>
Rev
Date:
Tuesday, November 18, 2025
Sheet
8
of
11

日期	项目	备注	后期修改点
2024.3.1	红外激光学	首版v1.00	无

Title <Title>		
Size A	Document Number <Doc>	Rev <RevCo
Date: Tuesday, November 18, 2025	Sheet 10 of 11	

Title	<Title>		
Size B	Document Number <Doc>		Rev <RevCode>
Date:	Tuesday, November 18, 2025	Sheet 11 of 11	